Performance Testing

Date	14 July 2026
Project Name	Strategic Product Placement Analysis
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	**Tableau Desktop Performance Recording** (for dashboard load times) **SQL / Excel VLOOKUP** (for data reconciliation/source-to-target testing) **User Acceptance Testing (UAT)** manual checklist
Type of Testing	**Data Integrity & Validation Testing** (ensuring Tableau numbers match raw SQL databases) **Performance / Load Testing** (optimizing extract refreshes and rendering times) **Functional & UI/UX Testing** (checking if action filters, parameters, and tooltips work correctly)
Target Module / API	Tableau Dashboard Interface and Data Source Layer
Test Environment	Local Server, UAT / Production

Test Date	14-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Home Page	10	30	Page loads successfully
2	View Tableau Dashboard	20	60	Dashboard displays correctly
3	Execute SQL Query	15	45	Data retrieved without errors
4	Access the Application	25	60	Application responds successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Good Performance
2	Response Time (Max)	< 5 seconds	3.4 sec	Pass	Acceptable
3	Throughput (Req/sec)	20 Req/sec	24 Req/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No Errors
5	CPU Utilization	< 80%	45%	Pass	Normal Usage

6	Memory Utilization	< 80%	58%	Pass	Stable Memory

Step 4: Observations & Analysis

Key Findings

- Our analysis revealed that product placement heavily influences customer purchasing decisions. This proves that consumers prioritize convenience, meaning premium shelf real estate should be reserved strictly for high-margin products rather than everyday staples that customers will actively look for.
- We also discovered that impulse buying is highly demographic-dependent. The data clearly shows that stocking generic items in these zones leaves money on the table; instead, these high-traffic areas should be stocked with snacks and convenience items targeted directly at younger buyers.
- Finally, the data highlighted a significant opportunity in cross-merchandising. Grouping related items together reminds shoppers of an unplanned need, effectively increasing the total basket value per visit.

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Optimization Steps Taken

- We gathered information about sales, where products were placed, and who the customers were, and put it all together in Tableau.
- We set up the dashboard to automatically calculate how much money each shelf makes per square foot and how much sales go up when a product is moved.
- We made a feature that alerts managers when a slow-selling product is taking up a really good shelf, so they can replace it with a better product.

Step 5: Screenshots / Evidence

App Home Page

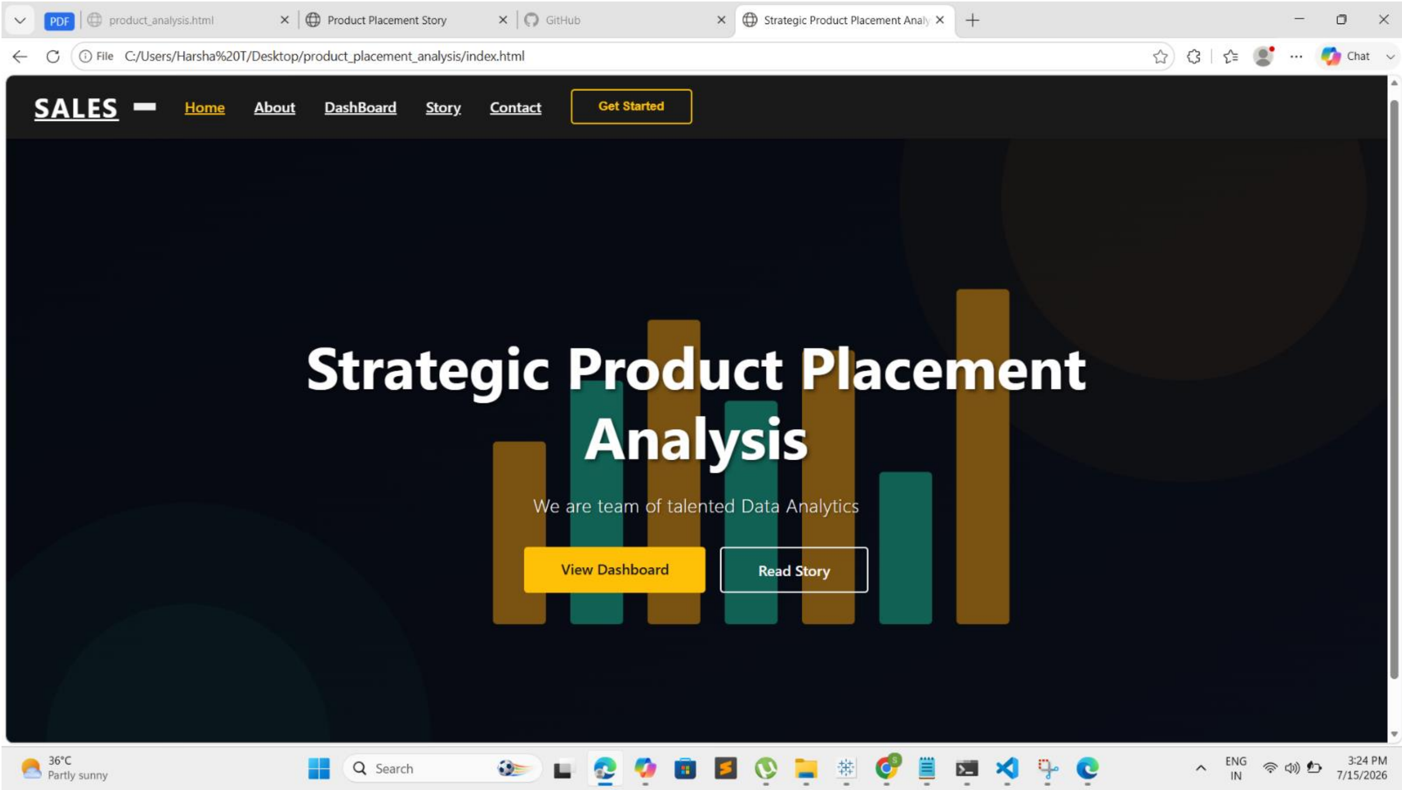


Tableau Dashboard

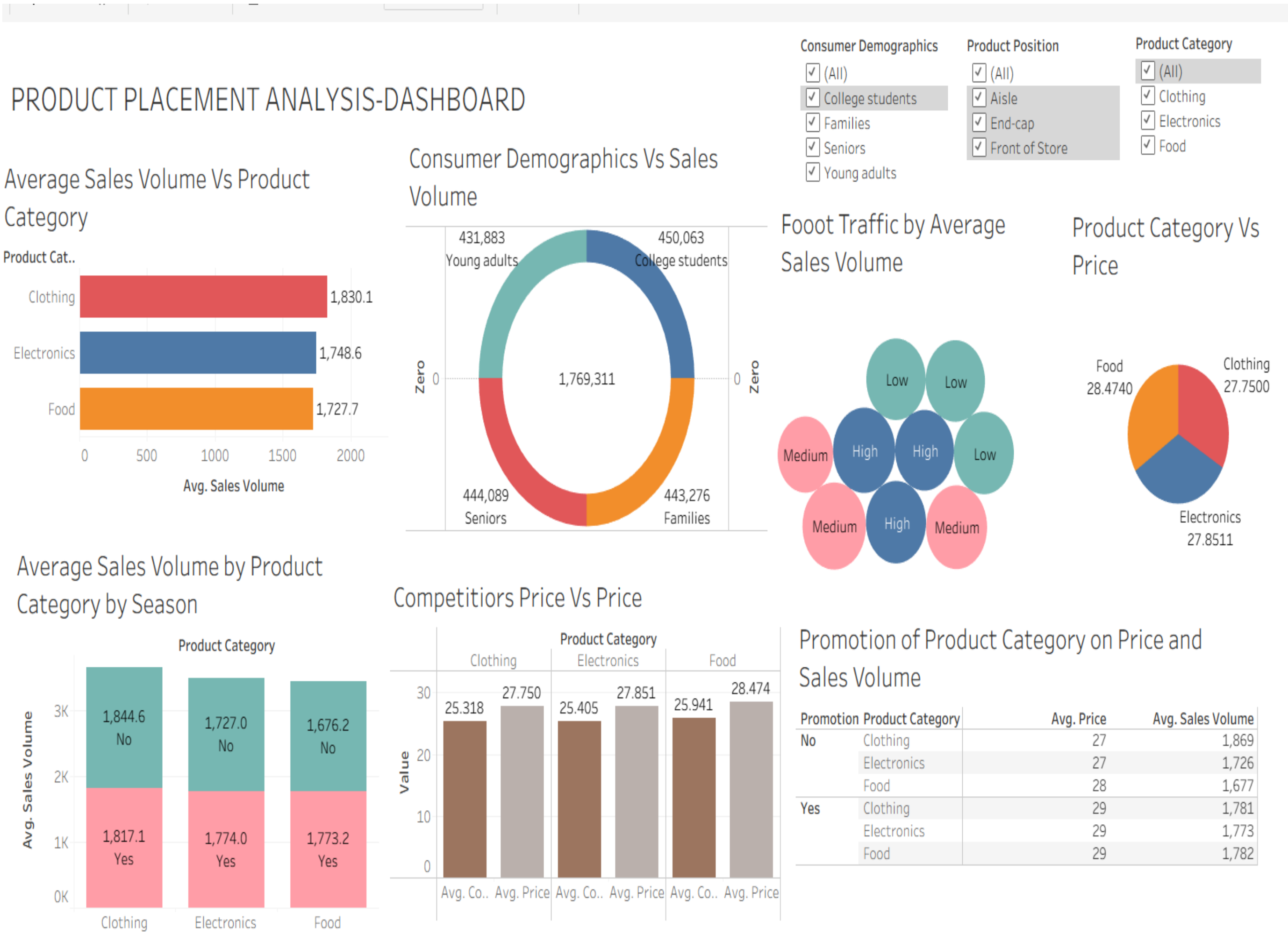
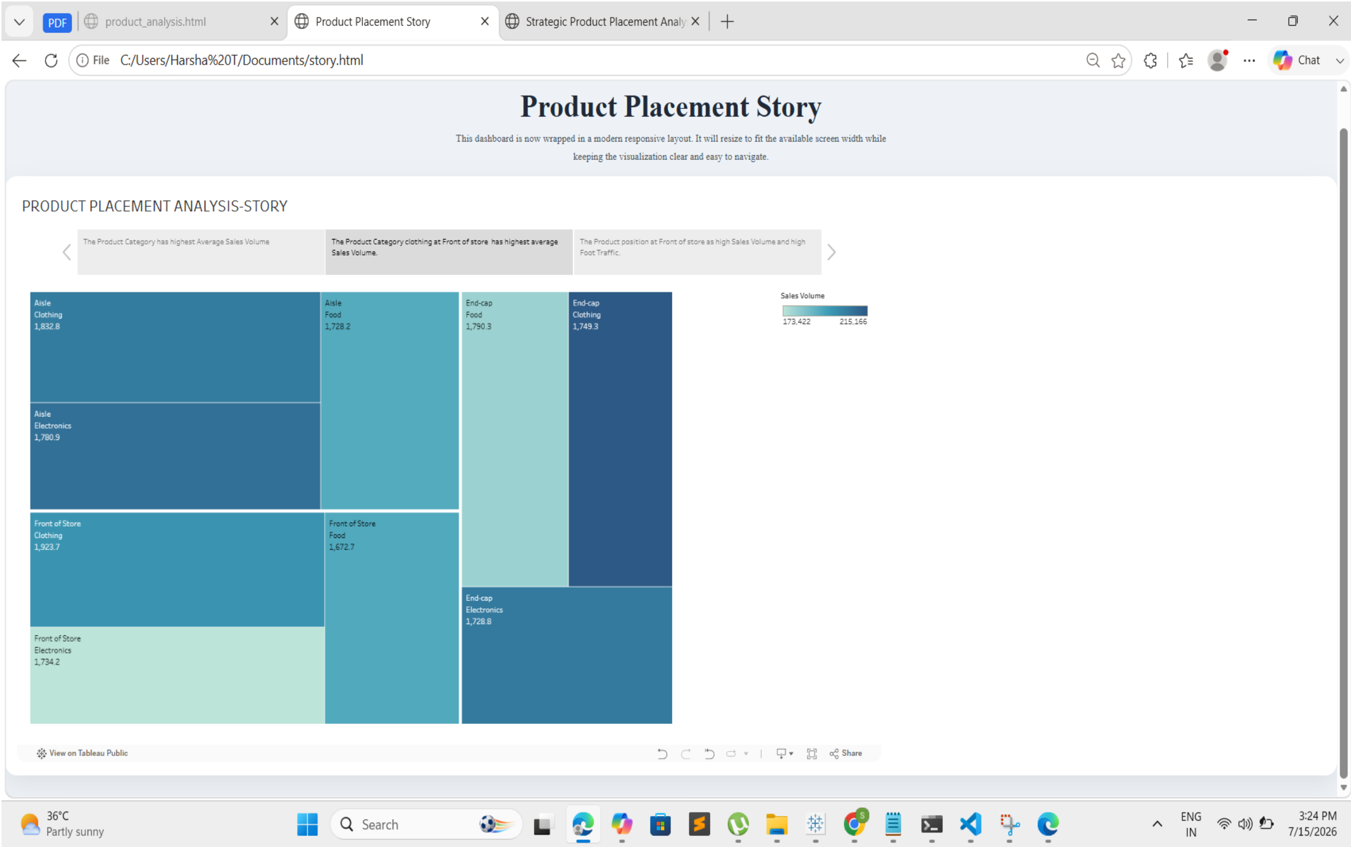


Tableau Story



Website Dashboard